E-INK DESK DISPLAY

Tablo

Your day, your classes and your home on one calm screen. No backlight, no notifications. Hang it on the wall or stand it on the desk, and it always shows what matters.

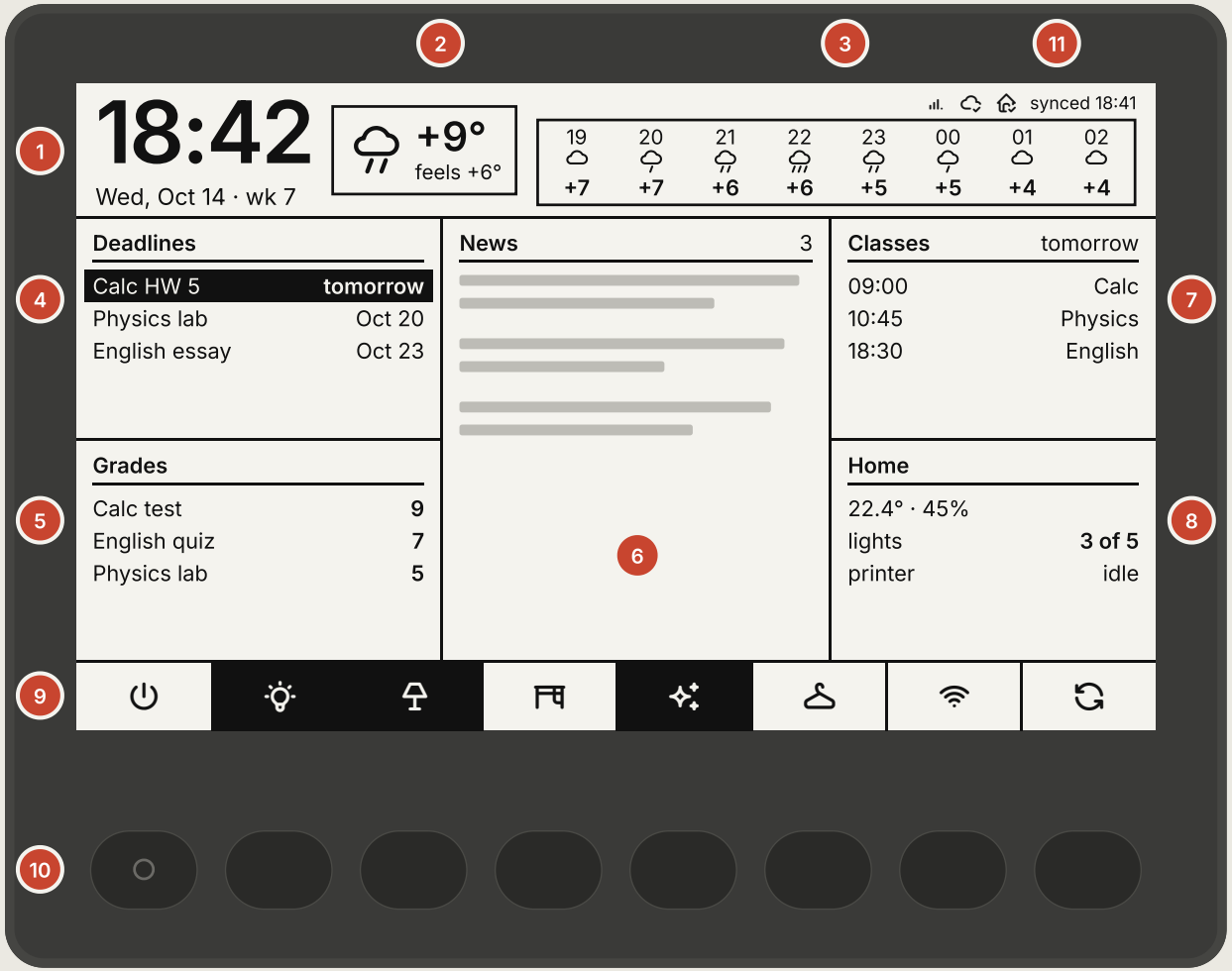
1. **Everything at a glance**
   Time, hourly weather, deadlines, news, grades, classes and home status.

2. **8 buttons for your home**
   Lights, scenes, anything in Home Assistant. Reassign them without reflashing.

3. **Paper, not a monitor**
   7.5″ electronic ink: no glow, no flicker, reads like a printout.



Concept · October 2026 · on-screen data is sample content



THE SCREEN

What goes where

Right now on top, what's coming in the middle, home controls at the bottom.

1. **Time and date.** Large, updated every minute without flashing.
2. **Weather now.** Icon, temperature, feels-like.
3. **Next 8 hours.** Icon and temperature per hour.
4. **Deadlines.** Upcoming due dates, urgent ones inverted.
5. **Grades.** Latest marks on a 10-point scale.
6. **News.** A short digest of the day.
7. **Classes.** Next schedule; tomorrow's in the evening.
8. **Home.** Temperature, humidity, lights on.
9. **Button icons.** Filled tile means it's on.
10. **Buttons.** Eight keys, each right under its icon.
11. **Status line.** Wi-Fi, server, Home Assistant and last sync; turns into a black “offline” tag if anything drops.

On the wall or on the desk

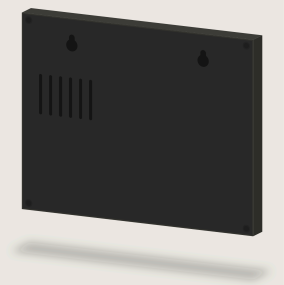
A flat 185 × 146 × 15 mm printed body hangs on two keyholes like a picture frame. For the desk, a slim leg clips into the same keyholes and holds it at 15°, hidden behind the screen. USB-C and the power switch sit on the right edge.



On the wall
hangs on two screws, cable straight down



On the desk
clip-on rear leg, invisible from the front



Back
keyholes, vents over the board



Right edge
USB-C and power switch

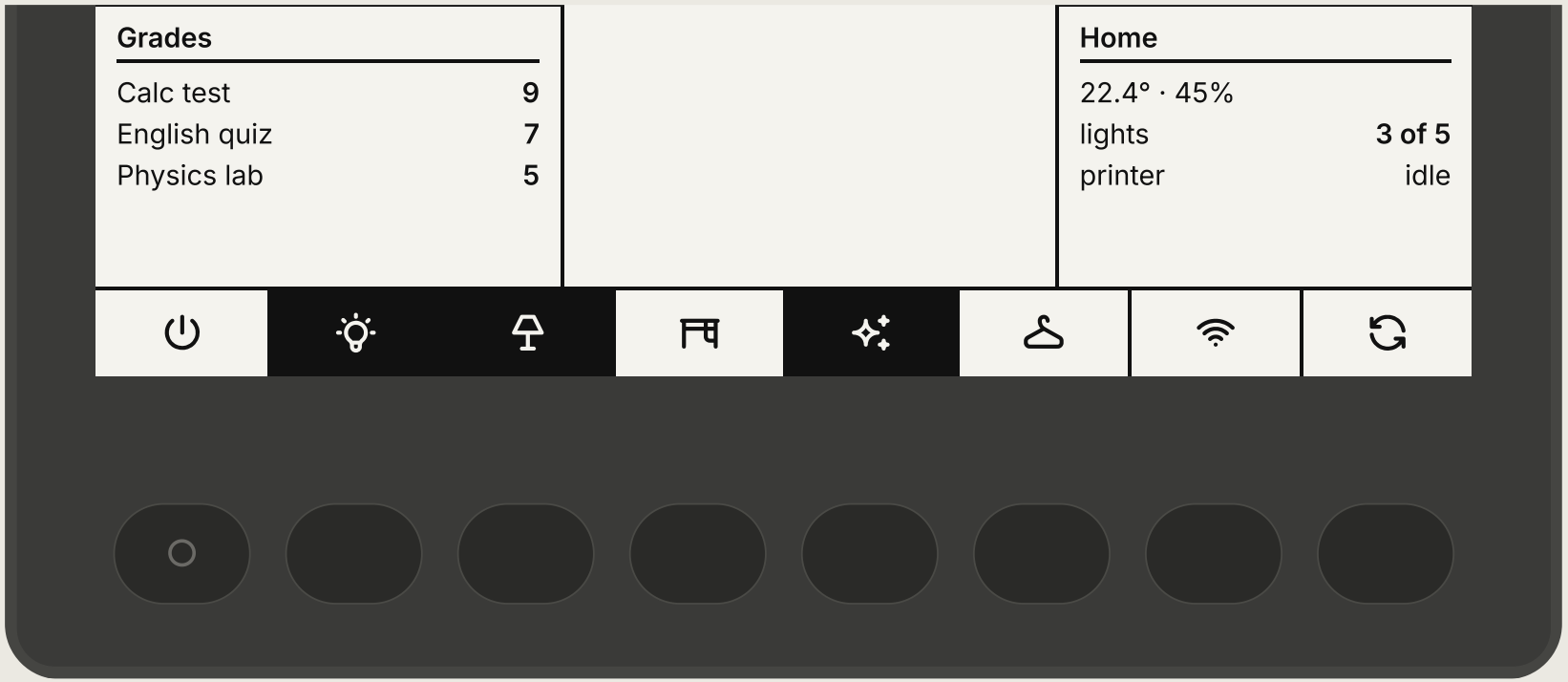


Side
15 mm body, 15° on the leg

BUTTONS

Eight buttons that know what they do

A row of low-profile mechanical switches under the screen. Each key sits exactly under its icon, 20.4 mm apart across the full width of the display.



Icons on the screen

What each key does is set in Home Assistant. Reassign it and the icon updates on its own.

See what's on

A filled tile means that light is on. It redraws in under a second after a press, without flashing the screen.

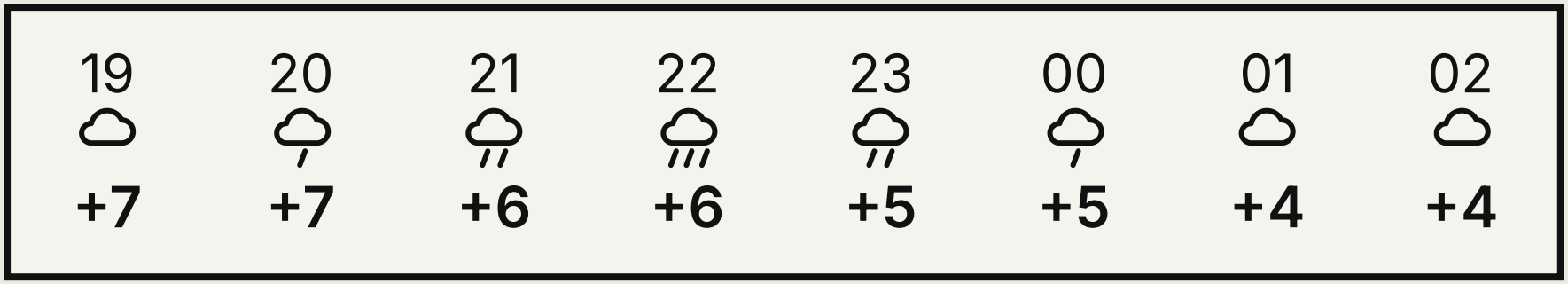
Pills you can feel

Keycaps are 3D-printed to match the case. "All off" has a dimple so you can find it without looking.

WEATHER

The next eight hours in one line

Each hour gets an icon and a temperature. Rain strength shows as one, two or three strokes under the cloud, with no percentages or charts. Data comes from Open-Meteo, free and keyless.



clear · clear night · cloudy · light rain · moderate · heavy

Only when it matters

Rain is shown only when it's actually likely, so a hesitant forecast doesn't look like a downpour.

One style

The large icon in the header and the small hourly ones share the same line and sit on the same baseline.

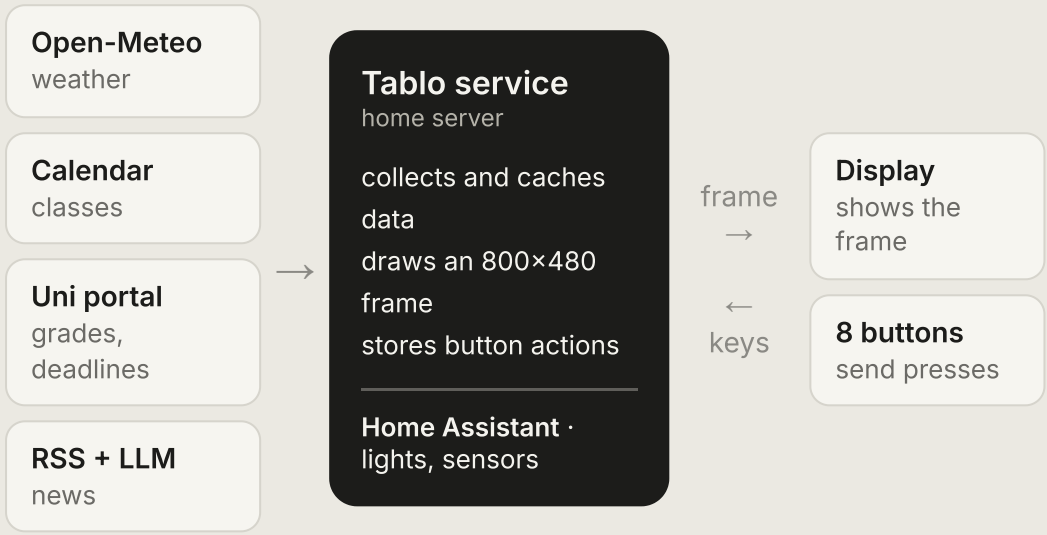
Easy to extend

Snow, storms, fog: the service maps the forecast's weather code to the right icon.

HOW IT WORKS

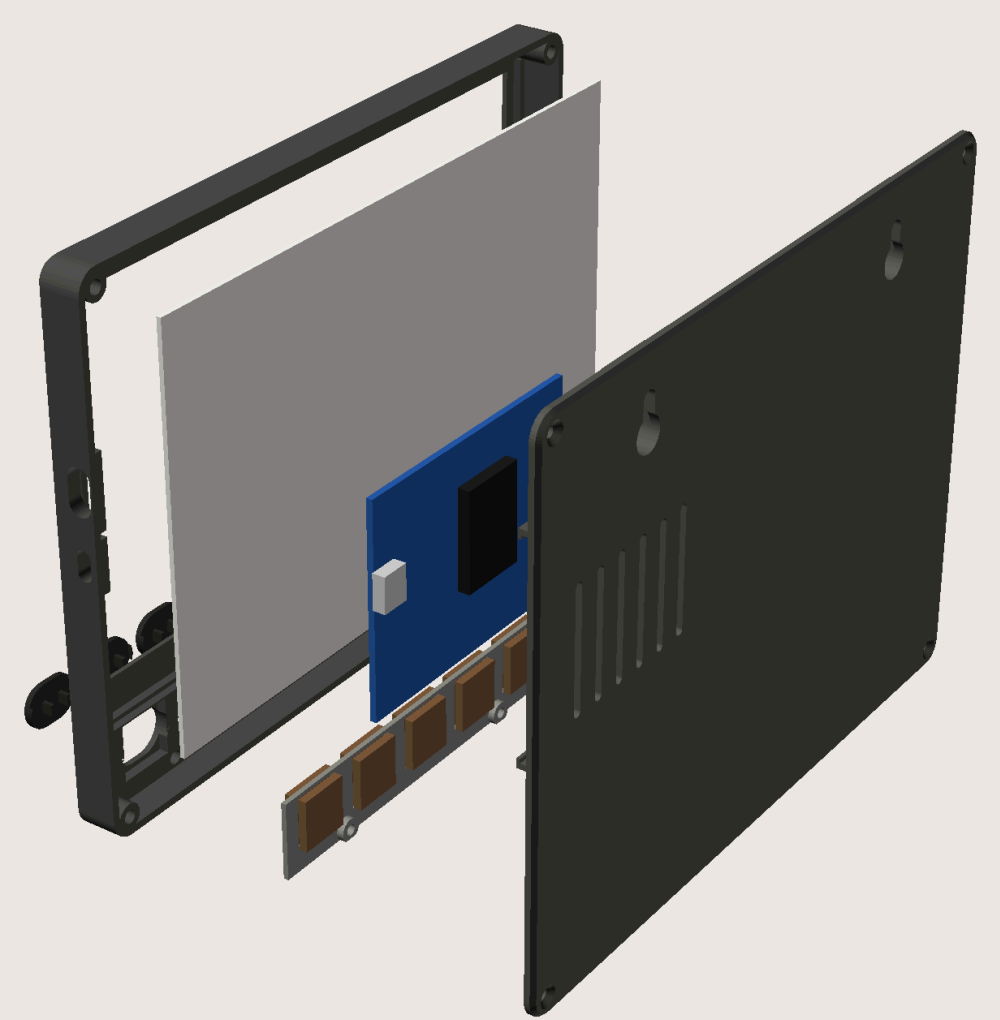
Smart server, simple screen

All the heavy lifting happens in a small service on the home server next to Home Assistant: it collects the data, draws a ready-made frame and handles button presses. The device just shows the picture and reports its keys.



SPECS

Display	7.5″ e-ink, 800×480, black and white, landscape Good Display GDEY075T7
Refresh	partial 0.3 s, full 3 s; full refresh hourly or overnight
Controller	Good Display ESP32-L + DESPI-C02 adapter, Wi-Fi
Buttons	8 × Kailh Choc v1 Brown, tactile 3D-printed pill keycaps, 20.4 mm pitch
Enclosure	flat 3D-printed body, 185 × 146 × 15 mm wall keyholes + clip-on rear leg for 15°
Power	USB-C on the right edge, always on
Fasteners	7 × M2.5 heat-set inserts, 4 × M2.5×8, 3 × M2.5×6
Integrations	Home Assistant, Open-Meteo, calendar, RSS



INSIDE

Five printed parts around real components

Everything is designed around the real parts: the panel sits on rails behind the bezel, switches clip into a separate plate, and the board snaps into the back cover with its USB-C port at the side wall.

Printed	front shell, switch plate, back cover, rear leg, 8 pill keycaps
Electronics	GDEY075T7 panel, ESP32-L board, DESPI-C02 adapter, 8 × Kailh Choc v1
Fit check	a 2-key test coupon is printed first to tune switch and keycap fit
Model	parametric OpenSCAD, every dimension in one file